

Segmented worms

PHYLUM	Annelida
CLASSES	4
ORDERS	17
FAMILIES	130
SPECIES	About 18,000

The bodies of segmented worms—or annelids—differ from other worms in being divided into a series of linked but partly independent sections, each segment containing the same set of organs. The segments are filled with fluid which, combined with the musculature, gives the body its shape. Segmented worms have various modes of life,

many are burrowers, living either in soil or in sediments beneath freshwater or the seawater.

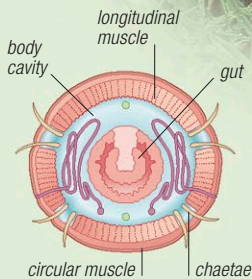
Anatomy

A segmented worm's body consists of a head at one end, a tail at the other, and many trunk segments in between. Each segment is divided from the others by partitions and contains distinctive organs. The mouth is on the first segment, and leads to the gut which passes through all the segments to end in an anus in the tail. On the side of each segment are hooks and bristles (chaetae). Some marine worms have sensory tentacles; others, such as the earthworms, have no obvious projections.



FRESHWATER PARASITES

Leeches are parasitic worms. They use 2 suckers, one at each end of the body, to attach themselves to their hosts.



BODY SECTION

The body cavity of an earthworm is filled with fluid and contains a large gut. Short chaetae project through 2 layers of muscle to the body's outer surface.

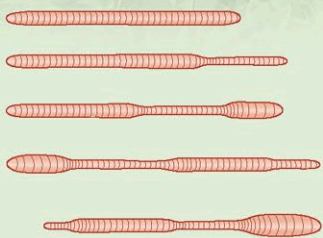
EARTHWORM BURROWS

Feeding mainly on decaying vegetable matter, earthworms are present in soils throughout the world. They are usually found close to the surface but rarely leave their burrows, to avoid being eaten by birds and other animals.



Movement

An earthworm moves by coordinating the action of muscles in different parts of its body. In each segment, there is one layer of longitudinal muscle and another of circular muscle. When contracted, these muscles make the segment either long and thin or short and fat. As the worm moves through its burrow, short, fat segments anchor part of the body against the wall, while long, thin segments penetrate farther through the soil.



WAVE MOVEMENT

When an earthworm moves, its segments alternately narrow and widen, forming waves of muscular contractions that pass along the body.

INVERTEBRATES

Class Polychaeta

Bristleworms

Occurrence 12,220 spp. worldwide; in burrows and tubes, in coral reefs, and amid seaweed and animal colonies on the seabed



Bristleworms, also called polychaetes, form the biggest and most varied class of segmented worms, ranging from less than 1/32 in (1 mm) to 9 3/4 ft (3 m) long. Lugworms, bloodworms and ragworms, tubeworms, and bamboo worms are all

members of this class. Despite being abundant and sometimes brilliantly colored, bristleworms often escape attention because of their burrowing or tube-dwelling lifestyles and their quick reactions when disturbed. The exceptions to this rule are the active hunters, such as ragworms, which can sometimes be spotted in rock pools or under stones. Most bristleworms are long, slender, and worm shaped, but some may be almost circular. They have parapodia—flaps bearing bristles, or chaetae—or at least ridges projecting from the sides of each segment, and heads that bear appendages. Two general types occur: free living and often carnivorous species, which have well-developed parapodia along the body; and sedentary species, which dwell in burrows or tubes, are deposit

or suspension feeders, and have very specialized parapodia. A few species are parasitic. Most bristleworms are solitary animals, but some tubeworms may form large colonies or reefs. The sexes are usually separate.

SEA MOUSE

Somewhat inappropriately named after the Greek goddess of love, *Aphrodita aculeata* has a feltlike mass of chaetae covering its dorsal surface and scales. This marine scaleworm is up to 8 in (20 cm) long, and is widespread.



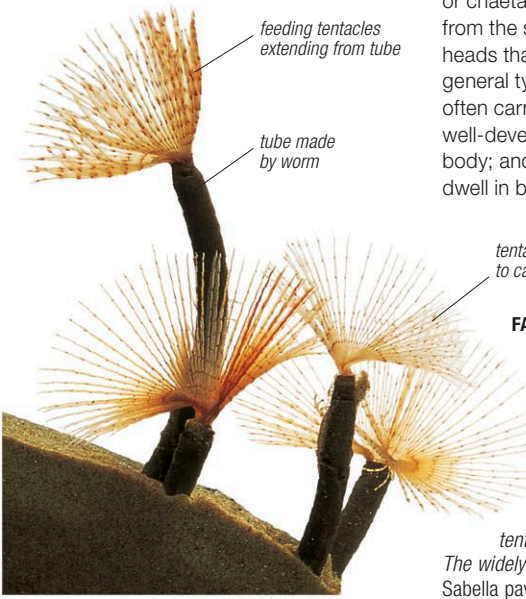
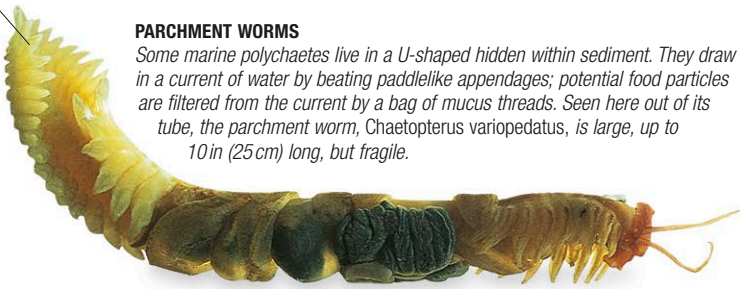
RAGWORMS

Many marine polychaetes, like the ragworm, live in burrows in soft sediment, but can swim well and actively pursue their prey. The predatory lifestyle of the Atlantic species *Alitta virens* (right)—which may be up to 20 in (50 cm) long—is reflected in the numerous sensory structures, including eyes and large jaws, at its head end.



PARCHMENT WORMS

Some marine polychaetes live in a U-shaped hidden within sediment. They draw in a current of water by beating paddlelike appendages; potential food particles are filtered from the current by a bag of mucus threads. Seen here out of its tube, the parchment worm, *Chaetopterus variopedatus*, is large, up to 10 in (25 cm) long, but fragile.



FANWORMS

Some polychaetes live within a tube of their own making, which may be attached to a hard surface or, as in this fanworm, partially buried in soft sediment. Their movement is largely restricted to within the tube, but their feeding tentacles extend out into the water. The widely distributed species shown here—*Sabella pavonina*—is 8 in (20 cm) long.